

Mark (K) Tinaga

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PROFESSIONAL SUMMARY

An engineer with extensive experience designing, writing and scaling high-performance software and real-time ML pipelines to handle large amounts of data. Experienced in building production-ready systems and ML models, leveraging advanced techniques and frameworks. Proven ability in enhancing system reliability, optimizing data processes, and collaborating with cross-functional teams to drive business value.

TECHNICAL SKILLS

Programming Languages: Python, JavaScript, C, Bash

Machine Learning Tools: TensorFlow, PyTorch, scikit-learn, Pandas, Iguazio

AI Tools & Technologies: GitHub Copilot, LLMs, RAG, Model Context Protocol (MCP), Prompt Engineering

Backend Frameworks: Django, Flask, FastAPI

Databases: PostgreSQL, MySQL, MongoDB, Redis

DevOps and Tools: Docker, Kubernetes, AWS, Jenkins, GitHub Actions

Testing and Quality: Unit Testing, TDD, Integration Testing, CI/CD/CT

Architecture and Design: REST APIs, Microservices, Event-Driven Design

Security: OAuth2, JWT, RBAC, API Security

Version Control: Git, GitHub

EXPERIENCE

Machine Learning Engineer

Mar 2025 - Present

Safaricom PLC, Nairobi, Kenya (Hybrid)

- Developed real-time ML pipelines on AWS (SageMaker, S3, EC2) and on-premise using Python, PySpark, and Kafka to process over 5,000 transactions per second, supporting 50 million customers
- Collaborated with cross-functional teams to design and implement feature engineering pipelines for large-scale ML model experimentation and deployment, handling over 10 TB of data weekly
- Integrated LLM and RAG-based approaches into internal tooling to improve feature discovery and retrieval across the team's growing feature store of 10,000+ features
- Engineered scalable data workflows with automated orchestration and CI/CD integration using Python and Jenkins, reducing data processing time by 30%
- Deployed and served ML models using MLRun and Iguazio, leveraging Nuclio serverless functions and Kubernetes pods to deliver low-latency, scalable model inference in production
- Adopted GitHub Copilot as a core part of the development workflow, accelerating pipeline development and improving code consistency across the team

Software/Machine Learning Engineer - Freelance

Dec 2019 - Present

Freelancer, Remote

- Designed and integrated APIs, including a custom SMS service using Africa's Talking API and a real-time notification system, bolstering user engagement by 15% and handling over 50,000 notifications daily
- Implemented continuous integration and deployment pipelines with GitHub Actions, automating tests and streamlining client project deployments, reducing deployment time by 40%
- Developed and deployed machine learning models using MLOps best practices with Python, TensorFlow, and MLflow, cutting model deployment time for client applications by 80%

Software Engineer - Fintech

Dec 2023 - Dec 2024

Huawei Technologies, Nairobi, Kenya

- Designed and deployed scalable architectures for mobile money and digital finance platforms (VOOMA, M-PESA, Cashia, ZIDII) serving over 30 million active users and processing billions of transactions annually
- Improved project efficiency through implementation of Acceptance Test Plans to validate solutions designed, cutting validation time by 25%
- Reduced integration errors by 20% by implementing robust architectural patterns and documentation

- Drafted functional requirement specifications (FRS), configuration requirement documents, and Acceptance Test Plans for over 10 major system functionalities

Software Service Engineer - Infrastructure

Jul 2023 - Dec 2023

Huawei Technologies, Nairobi, Kenya

- Maintained critical overdraft system with 99.999% uptime through proactive monitoring and optimization, supporting over 1 million daily users
- Implemented automated monitoring and alerting systems for early incident detection, reducing mean time to detection (MTTD) by 30%
- Enhanced system security through regular updates and threat analysis, closing over 10 critical vulnerabilities

Backend Software Engineer (Part-time)

Sep 2023 - Sep 2024

Zuyu, London, UK (Remote)

- Developed secure authentication APIs using MERN (MongoDB, Express.js, React and Node.js) stack with TypeScript to handle over 1 million requests per day
- Implemented role-based access control system enhancing platform security and improving compliance, reducing unauthorized access attempts by 25%
- Optimized API performance through caching strategies and database query optimization, reducing average API response time by 30%
- Collaborated with the frontend team to ensure seamless integration with the backend authentication system, reducing integration-related bugs by 15%

Data Analyst (Intern)

Jun 2021 - Aug 2021

Shining Hope For Communities (SHOFCO), Nairobi, Kenya

- Automated data processing and reporting using Python, generating over 300 reports to optimize COVID-19 vaccination strategies, saving approximately 20 hours of manual work per week
- Analyzed data trends to support decision-making, contributing to a 2% increase in vaccination rates, leading to over 1,500 additional vaccinations
- Streamlined report generation through process automation, reducing report delivery time by 50%

EDUCATION

Software Engineering Bootcamp

Aug 2022 - Aug 2023

ALX-Africa (Holberton School), Nairobi, Kenya

- Intensive 12-month software engineering program (70 hours/week)
- Focus: Low-Level Programming, High-Level Programming, DevOps

BSc. Mathematics and Computer Science

Sep 2018 - Jul 2023

Kenyatta University, Nairobi, Kenya

- Major: Applied Mathematics and Computer Science

Relevant Coursework:

- Data Structures and Algorithms
- Probability and Statistics
- Linear Algebra
- Numerical Analysis
- Discrete Mathematics
- Object-Oriented Programming
- Database Systems
- Computer Networks
- Human-Computer Interaction
- Neural Networks

ACADEMIC INTERESTS

Artificial Intelligence & Machine Learning: Neural Networks and Deep Learning, Distributed ML Systems, AI Security and Privacy

Systems Programming: Low-level System Design, Operating Systems, Computer Architecture

Algorithms & Data Structures: Algorithm Analysis and Design, Advanced Data Structures, Computational Complexity